Atlas AI

About Atlas AI agents

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Learn how Atlas AI agents use language models, knowledge graphs, prompts, and tools to solve industrial business problems.

Atlas AI agents use generative AI (GenAI) language models to solve **specific** industrial business problems, like providing insights into historical and planned maintenance and analysis of time series data for root cause analysis.

They include prompts with **instructions**, industry-relevant tools, and access to industrial data stored in the Cognite Data Fusion (CDF) knowledge graph.

Agents interact with humans to meet predetermined objectives. Humans provide instructions, and the AI agent suggests the best actions to take to achieve those objectives.

Language models

Language models are one of the main ingredients of industrial AI agents. They're probabilistic models designed to **understand** and **generate** human language. **Large language models** (LLMs) are trained on vast amounts of data, making them versatile and useful for **general purposes**.

The **Language model library** has detailed reference information about the available language models to help you choose.

Knowledge graphs

Cognite Data Fusion (CDF) has **data modeling features** and **out-of-the-box data models** to get you started building a structured, flexible, and contextualized **knowledge graph** of your

tasks such as data analysis and problem-solving. The decisions are **traceable** and **transparent** so that you can always see how the AI agent arrived at its answers.

Industrial data is not static, and the knowledge graph ensures that the language models always work with the **most recent** data. **Access control** is another advantage of using knowledge graphs, controlling what data is provided to the language model.

Language models and knowledge graphs **complement** each other, and the strengths of one technology compensate for the limitations of the other.

For guidance on making your data model agent-ready, see **Optimizing data models for AI**.

Prompts and prompt engineering

Prompts—prompt engineering—is how you optimize the output from the agents by helping the language models understand what output you want them to produce. **Instructions** are crucial for effectively guiding AI models.

Instructions are the specific directives and guidelines provided to the agent that define both what you want to achieve and how to achieve it. They detail the desired outcomes, objectives, and the specific approach the language model should take. For instance, an instruction might specify that you want to generate a summary, define the format of that summary, and include the key points to cover.

Instructions should clearly outline both **what** you want to accomplish and **how** to get there.

Learn more

- Amazon Bedrock: **Prompt engineering concepts**
- Open AI/ChatGPT: **Prompt engineering**
- Anthropic/Claude: **How to prompt engineer**
- Google/Gemini: **Introduction to prompt design**
- Microsoft/ChatGPT: **Prompt engineering techniques**

Atlas AI agents can interact with tools and functions, allowing the agent to perform complex tasks or interact with your application. We provide built-in tools for Atlas AI agents, but you can also define your own tools to extend their capabilities.

Typically, tools fall into these broad categories:

- **Querying and exploring data** from the industrial knowledge graph ensuring that the information is relevant and task-specific.
- **Running code** and **algorithms**, or perform **computations** in real time based on input or operational requirements.
- **Integration with external systems** via APIs to pull data, trigger workflows, or update records.
- **Integration with collaboration and communication tools**, for example, Slack and Microsoft Teams, ensuring seamless human-machine interaction.

To view the currently available tools, see the **Agent tools library**.

Runtime versions

Runtime versions define the capabilities and behavior of your Atlas AI agents. Each version includes a specific system prompt, a set of agent tools, and a defined feature set. When you create an agent, Atlas AI automatically pins it to the latest stable runtime version.

The agent uses a specific runtime version to provide **stability** for production deployments. Cognite regularly releases improvements to Atlas AI, but your agent maintains **consistent behavior** with a pinned runtime version. You control when to upgrade your agent to a newer runtime version.

For detailed information about available versions and their features, see **Agent runtime versions**.

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